On April 24 we'll start using GitHub Copilot interaction data for AI model training unless you opt out. Review this update and manage your preferences in your GitHub account settings.

ECE 157C / 272C — Homework 1 Report

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1. Setup and Execution

Dependencies

```
pip install langgraph langchain langchain-openai pandas python-dotenv
```

Configuration

1. Create a `.env` file in the project root with your OpenAI API key:

   ```
   OPENAI_API_KEY=sk-...
   ```

2. The agent uses **GPT-4o** via the OpenAI API (`langchain-openai`).

How to Run

```
# Run all 18 questions (9 housing + 9 custom) and generate results.csv
python3 test_agent.py
```

The script will:

- Run each question through the LangGraph agent
- Print evaluation (PASS/FAIL) and a preview of each answer
- Write all results to `results.csv`

Project Structure

```
project/
├── agent.py              # LangGraph workflow definition
├── nodes.py              # Node functions (codegen, execute, evaluate, retry, respond)
├── test_agent.py         # run_agent() function + question runner
├── housing.csv           # Fixed dataset (California Housing, 10,000 rows)
├── custom_dataset.csv    # Custom dataset (Video Game Sales, 10,000 rows)
├── results.csv           # Generated results for all 18 questions
└── Report.pdf            # This report
```

2. Prompt Log (Development Prompts)

Below is a log of prompts used with the AI coding assistant (GitHub Copilot / Claude) during development.

Prompt 1: Initial Project Scaffolding

Goal: Generate the initial project structure with all required files.

Prompt (paraphrased):

> "Read through HW1 and come up with a plan to complete the HW. Help me implement, let's use OpenAI because I have a key."

Outcome: The assistant created `nodes.py`, `agent.py`, and `test_agent.py` with the full LangGraph pipeline. The initial code structure was functional on the first attempt.

Prompt 2: Fixing Environment Loading

Goal: Fix `OPENAI_API_KEY` not being loaded when running `test_agent.py` directly.

Prompt (paraphrased):

> "Run all 9 housing questions."

Issue: The `nodes.py` module-level `ChatOpenAI()` instantiation failed because `load_dotenv()` was only called in `agent.py`, which imported from `nodes.py` — so the key wasn't loaded in time.

Fix: Added `load_dotenv()` at the top of `nodes.py` before the `ChatOpenAI` instantiation.

Outcome: Helpful — resolved the import ordering issue immediately.

Prompt 3: Custom Dataset Setup

Goal: Download and configure the Video Game Sales dataset as the custom dataset.

Prompt (paraphrased):

> "Let's do video games sales."

Outcome: The assistant used `kagglehub` to download the dataset, sampled it to 10,000 rows, and wrote 9 questions at three difficulty levels. Helpful.

Prompt 4: Fixing Advanced Question Failures

Goal: Debug why 4 out of 9 custom dataset questions failed.

Prompt (paraphrased):

> Iterative debugging — inspected generated code, identified that the LLM was guessing wrong column names (e.g., `Japan_Sales` instead of `JP_Sales`).

Fix: Added a `_get_csv_info()` helper that reads the first few rows of the CSV and injects column names, dtypes, and sample data into the codegen prompt. Also added instructions: "Use ONLY the exact column names listed above."

Outcome: Very helpful — all 4 failing questions started passing after this change.

Prompt 5: Improving Execute Node Robustness

Goal: Detect when generated code fails to assign a value to `result` or produces an empty DataFrame.

Fix: Enhanced `execute_node` to check for `None` result and empty DataFrames/Series, and return descriptive error messages to guide the retry.

Outcome: The retry node now receives actionable feedback, improving code correction on the second attempt.

Prompt 6: Improving Evaluator Prompt

Goal: Fix overly strict evaluator that marked a correct "top 5 best-selling games" answer as FAIL.

Fix: Changed evaluator prompt from "correctly and fully answers" to "produced a non-empty, reasonable result that addresses the question." FAIL is now reserved for empty, None, error, or completely unrelated results.

Outcome: Reduced false FAIL evaluations while still catching genuine failures.

3. Failure Analysis and Debugging

Failure Case 1: Wrong Column Names (Custom Dataset — Advanced Questions)

Question: "Identify genres that perform disproportionately well in Japan compared to North America."

Did the code work on the first attempt? No.

What went wrong: The LLM generated code referencing `Japan_Sales` and `North_America_Sales`, but the actual column names are `JP_Sales` and `NA_Sales`. The code raised a `KeyError`, the retry also guessed wrong because it had no column information.

Debugging: Printed the generated code and saw `print(df.columns)` statements — the LLM was trying to discover columns at runtime instead of using the correct names. The root cause was that the codegen prompt did not include any dataset schema information.

Prompt improvement: Added `_get_csv_info()` to read actual column names, dtypes, and sample rows from the CSV, and injected this into both the codegen and retry prompts. Added the rule: "Use ONLY the exact column names listed above. Do NOT guess column names."

Did the revised prompt improve the result? Yes — all column-related errors were eliminated. The question passed on the next run.

Failure Case 2: Empty Result Variable (Custom Dataset — Platform Lifecycle)

Question: "Find platforms that had a rapid rise and decline in game releases. What does this suggest about the console lifecycle?"

Did the code work on the first attempt? No.

What went wrong: The generated code referenced `Year_of_Release` instead of `Year`. The `dropna(subset=['Year_of_Release'])` call raised a `KeyError`. On retry, the code was fixed but used filtering logic that resulted in an empty DataFrame assigned to `result`. The evaluator saw `None` and returned FAIL, but the respond node still generated a "no data found" answer.

Debugging: Inspected the execution result and found it was `None`. The execute node was setting `result = env.get("result", "No 'result' variable found.")` — which returned a string even when the actual `result` variable was an empty DataFrame. Enhanced the execute node to explicitly check for empty DataFrames/Series and report them as errors.

Prompt improvement: Combined the column-name fix (Failure Case 1) with the empty-result detection. The retry prompt now receives a clear error message: "The `result` variable is an empty DataFrame/Series. Adjust the code logic so `result` contains meaningful data."

Did the revised prompt improve the result? Yes — the retry generated code that used correct column names and logic, producing a non-empty result. The question passed.

4. Self-Grading

#	Dataset	Difficulty	Question	Eval	Correct?
1	housing	Simple	Average median house value	PASS	Yes — $206,943.86 matches manual calculation
2	housing	Simple	Highest avg by ocean proximity	PASS	Yes — ISLAND is correct
3	housing	Simple	Min, max, median of house value	PASS	Yes — 14,999 / 500,001 / 178,750
4	housing	Intermediate	Median income by ocean proximity	PASS	Yes — values are reasonable
5	housing	Intermediate	Geographic areas with highest prices	PASS	Yes — SF Bay Area region
6	housing	Intermediate	Population density vs house value	PASS	Yes — weak negative correlation (-0.044)
7	housing	Advanced	Top 5 expensive areas + factors	PASS	Yes — correct ranking and reasonable factors
8	housing	Advanced	Low-price coastal areas	PASS	Yes — identified areas below median with contributing factors
9	housing	Advanced	Similar income, different value	PASS	Yes — ocean proximity is a key differentiator
10	vgsales	Basic	Total global sales	PASS	Yes — 5,338.63M
11	vgsales	Basic	Platform with most games	PASS	Yes — DS
12	vgsales	Basic	Top 5 best-selling games	PASS	Yes — Wii Sports at #1
13	vgsales	Intermediate	Avg sales by genre	PASS	Yes — Platform and Shooter highest

#	Dataset	Difficulty	Question	Eval	Correct?
14	vgsales	Intermediate	Top publisher + regional breakdown	PASS	Yes — Nintendo
15	vgsales	Intermediate	Sales trend by year	PASS	Yes — peak around 2008-2009
16	vgsales	Advanced	JP vs NA genre performance	PASS	Yes — Role-Playing disproportionately high in JP
17	vgsales	Advanced	Platform rise and decline	PASS	Yes — identified volatile platforms
18	vgsales	Advanced	Publisher regional patterns	PASS	Yes — Ubisoft/Activision (NA) vs Enix/Namco (JP)

5. Advanced Question Reflection

Housing Dataset — Advanced Questions

Q7: Top 5 most expensive geographic areas

Is the answer the only possible correct answer? No. The agent grouped by `ocean_proximity` category, yielding 5 regions. An equally valid approach would be to bin by latitude/longitude to identify specific neighborhoods.

Can the question be interpreted differently? Yes — "geographic areas" is ambiguous. It could mean ocean proximity categories, lat/long grid cells, counties, or cities. Different interpretations yield different top-5 lists.

Could different answers be valid? Yes. Grouping by lat/long bins would identify specific neighborhoods (e.g., parts of San Francisco), while grouping by ocean proximity gives broader regional trends.

How could the question be rewritten? "Identify the top 5 most expensive latitude-longitude grid cells (using 1-degree bins) and explain the key factors contributing to their high prices."

Q8: Low-price coastal areas

Is the answer the only possible correct answer? No. The definition of "relatively low" is subjective. The agent used below-median as the threshold, but one could use the 25th percentile or a fixed dollar amount.

Can the question be interpreted differently? Yes — "coastal areas" could mean only `NEAR OCEAN`, or it could include `<1H OCEAN` and `NEAR BAY`. The agent included all three, which is a reasonable but not the only interpretation.

Could different answers be valid? Yes. A stricter definition of "coastal" (only `NEAR OCEAN`) or a different threshold for "relatively low" would produce different results.

How could the question be rewritten? "Among areas classified as NEAR OCEAN, find those with median house values below the 25th percentile. What factors (income, housing age, population) differ from higher-priced NEAR OCEAN areas?"

Q9: Similar income, different house value

Is the answer the only possible correct answer? No. The agent grouped by exact median income values, which is very granular. Binning income into ranges (e.g., $2-3, $3-4) would be more meaningful.

Can the question be interpreted differently? Yes — "similar" income levels could mean exact match, within $500, or within the same quartile. "Significantly different" house values is also subjective.

Could different answers be valid? Yes. Different binning strategies and thresholds would surface different area pairs.

How could the question be rewritten? "Group areas into median income bins of width 0.5. Within each bin, find the areas with the highest and lowest median house values. What non-income factors explain the gap?"

Video Game Sales Dataset — Advanced Questions

Q16: Genres disproportionately strong in Japan vs NA

Is the answer the only possible correct answer? No. The agent used a JP/NA sales ratio > 1 as the threshold. Using a different metric (e.g., percentage of total sales within each region) could surface additional genres.

Can the question be interpreted differently? Yes — "disproportionately well" could mean higher absolute sales, higher market share within the region, or a higher ratio relative to overall genre popularity.

Could different answers be valid? Yes. Role-Playing is the clearest answer, but depending on methodology, Puzzle or Strategy might also qualify.

How could the question be rewritten? "For each genre, calculate its share of total JP sales and its share of total NA sales. Which genres have a JP share at least 2x their NA share?"

Q17: Platforms with rapid rise and decline

Is the answer the only possible correct answer? No. The agent identified platforms with any peak-trough pattern, which includes nearly every platform. A more selective approach (e.g., requiring the peak to be at least 50 games and the decline to be >50%) would narrow the list.

Can the question be interpreted differently? Yes — "rapid" is subjective. It could mean within 2-3 years, or it could mean a percentage drop threshold. The agent used any year-over-year peak/decline.

Could different answers be valid? Yes. A more focused analysis might highlight only Wii, DS, and PS2 as the clearest examples.

How could the question be rewritten? "Identify platforms where annual game releases more than doubled in under 3 years and then dropped by more than 50% within the next 3 years. What does this suggest about console lifecycle duration?"

Q18: Publishers with regional sales disparities

Is the answer the only possible correct answer? No. The thresholds for "high" and "low" sales (1.0 and 0.1 respectively) are arbitrary. Different thresholds would surface different publishers.

Can the question be interpreted differently? Yes — "high" and "low" could be defined in absolute terms, relative to the publisher's total, or relative to the market average. The agent used per-game thresholds rather than aggregate publisher totals.

Could different answers be valid? Yes. Using aggregate publisher-level totals instead of per-game thresholds might highlight different publishers.

How could the question be rewritten? "For publishers with at least 20 games, rank by (total NA sales / total JP sales) ratio. List the top 5 NA-dominant and top 5 JP-dominant publishers. What genre and platform preferences differ between these groups?"